HARYANA LEET 2026 — FORMULA & QUICK REVISION SHEET

Section D: Other Engineering Streams | Questions 71 – 90 | 20 Marks

90 Questions · 90 Minutes · +1 / No Negative Marking · English Medium · Online MCQ

d-1 Civil (5Q)	d-2 Textile (5Q)	d-3 Chem/Print (2Q)	d-4 Ceramic (2Q)	d-5 Food Tech (2Q)	d-6 Others (4Q)

[d-1] CIVIL ENGINEERING

Building Materials — Bricks & Cement

- Bricks: clay+silt+sand, moulded and burnt at ~900–1000°C
- Standard brick size: 190×90×90 mm (with mortar: 200×100×100 mm)
- Cement types: OPC (gen use), PPC (pozzolana), SRC (sulphate resistant), RHC (rapid hardening), White cement
- ■ ***Grades of concrete: M10, M15, M20, M25, M30 (M=Mix, number=fck in N/mm^2)***
- Workability: water-cement ratio ↑ → workability ↑ but strength ↓
- Compaction: removes air voids; improves strength

Soil Mechanics

- Index properties: water content (w), void ratio (e), porosity (n), degree of saturation (S)
- ■ ***w = (mass of water)/(mass of solids) × 100%***
- ■ ***Void ratio $e = V_v/V_s$ | Porosity $n = V_v/V \times 100\%$***
- Atterberg limits: LL (liquid), PL (plastic), SL (shrinkage); PI = LL – PL
- ■ ***Darcy's law (seepage): q = kiA (k=permeability, i=hydraulic gradient, A=area)***

Foundation

- Shallow: spread (isolated pad), strip, raft/mat — when good soil near surface
- Deep: pile (end-bearing or friction), well, caisson — for soft/weak surface soil
- RCC basics: reinforcement carries tension; concrete carries compression
- ✔ **Cover to reinforcement: typically 25–40 mm for protection against corrosion**

Water Supply & Sanitation

- Per capita demand: domestic 135 lpcd; industrial & commercial extra
- Sources: surface (river, lake, reservoir) and ground (tube well, open well)
- Treatment sequence: Screening → Sedimentation → Coagulation → Filtration → Disinfection
- Chlorination: most common disinfection; residual chlorine 0.2 mg/L at consumer end
- Sludge disposal: digestion (anaerobic) → gas+stabilised sludge; drying beds, incineration

Surveying

- Chain surveying: only linear measurements; used for small flat areas
- Compass surveying: measures bearings; Prismatic compass (360° WCB system)
- Bench Mark (BM): reference point of known RL (reduced level)
- ■ ***RL (HI method): $HI = RL_{BM} + BS$ | $RL_{point} = HI – FS$***

Roads, Bridges & Earth Moving

- Flexible pavement: bitumen layers; load distributed; cheap but needs maintenance
- Rigid pavement: cement concrete; costly but durable; load distributed by slab action
- Road materials: WBM, DBM, BC (bituminous concrete), GSB (granular sub-base)
- Bridge types: beam, arch, suspension, cantilever, cable-stayed, truss
- Earth-moving machines: Dozer (push), Grader (level), Scraper (cut+carry), Excavator (dig)

Irrigation, Dams & Environment

- Irrigation methods: surface (flood, furrow, border), drip (micro), sprinkler
- Dam types: gravity (resists water by own weight), arch, buttress, earthen
- Site selection: narrow gorge, impermeable rock, seismically stable zone
- Water pollution indicators: BOD (biological), COD (chemical), DO (dissolved oxygen)
- Ecology: food chain, ecosystem, biodiversity; water pollution causes & remedies
- Non-conventional energy: solar (PV panels), wind (turbines), biogas, small hydro, tidal
- Rocks classification: Igneous (granite), Sedimentary (limestone, sandstone), Metamorphic (marble, slate)

[d-2] TEXTILE ENGINEERING

Fibre Classification

- Natural fibres: Cotton (cellulose, seed hair), Jute (bast), Wool/Silk (protein)
- Man-made: Viscose/Rayon (regenerated cellulose)
- Synthetic: Nylon (polyamide), Polyester, Acrylic, Spandex/Lycra
- Key properties: tenacity, elongation, moisture regain, thermal resistance, abrasion

Yarn Manufacturing

■ ***English count Ne = (length in hanks of 840 yd) / weight in pounds***

✔ **Higher count = finer yarn | Lower count = coarser yarn**

- Cotton spinning: Blow room→Carding→Drawing→Combing→Roving→Ring frame→Winding
- Open-end (rotor) spinning: faster, simpler, no roving stage

Fabric Manufacturing

- Weaving: interlacing warp (length) and weft (width) yarns on loom
- Basic weaves: Plain (1/1), Twill (2/2 or 3/1 — diagonal), Satin (float surface)
- Knitting: interlocking loops; warp knit (run-resist) vs weft knit (stretchable)

Wet Processing

- Scouring: removes natural impurities (wax, oils, size) using NaOH or soap
- Bleaching: whitening using H_2O_2 (safer) or NaOCl (optical)
- Mercerization: NaOH treatment of cotton → lustre, strength, dye affinity
- Dyeing methods: exhaust (batch), continuous (pad-dry-cure), semi-continuous
- Printing methods: Screen (flat/rotary), Block, Digital (inkjet), Transfer

Textile Testing

- Yarn: tensile strength (Instron), evenness (Uster tester), twist (twist tester)
- Fabric: tensile (grab test), tear (Elmendorf), bursting (diaphragm), pilling (ICI box), abrasion
- Quality control: acceptable quality level (AQL), statistical process control (SPC)

[d-3] CHEMICAL / PRINTING TECHNOLOGY

Chemical Engineering Basics

■ ***Reynolds No. $Re = \rho vD/\mu$ | $Re<2100$ laminar | $Re>4000$ turbulent***

■ ***Bernoulli + friction: $P_1/\rho g+v_1^2/2g+z_1 = P_2/\rho g+v_2^2/2g+z_2+h_f$***

- Chemical process industries: petrochemicals, fertilizers, pharmaceuticals, paint, dyes
- Agro-based industries: sugar, paper/pulp, vegetable oil, starch, alcohol (distilleries)
- Petro-chemicals: ethylene, propylene, benzene, ammonia — derived from crude oil/gas
- Unit operations: heat exchange, distillation, absorption, filtration, drying, evaporation

Printing Technology

- Letterpress: raised image area; oldest method; direct contact
- Offset lithography: image on flat plate; ink repelled by water; plate→blanket→paper
- Gravure/Intaglio: image engraved in cylinder; ink fills recesses; doctor blade wipes excess
- Screen/Stencil: ink pushed through mesh; used for textiles, posters
- Digital: Inkjet (drops of ink) or Laser (toner+heat); no plate required

✔ **Offset = most common for commercial printing; Digital = short runs, personalisation**

[d-4] CERAMIC ENGINEERING

- Ceramics: inorganic, non-metallic, formed at high temperatures (sintering/firing)
- Pottery types: Earthenware (porous, low temp ~1000°C), Stoneware (dense, ~1200°C), Porcelain/China (translucent, ~1400°C)
- Raw materials: Kaolinite ($Al_2Si_2O_5(OH)_4$), Feldspar (flux), Silica/Quartz (filler)
- Properties: very high melting point, hard, brittle, corrosion resistant, good insulator
- Processing: Batching → Mixing → Forming (slip casting, pressing, extrusion) → Drying → Firing → Glazing → Glaze firing
- Advanced ceramics: Al_2O_3 (alumina), SiC, Si_3N_4, ZrO_2 — cutting tools, aerospace, medical implants

■ ***Refractoriness: ability to withstand high temp without deforming; measured by PCE (Pyrometric Cone Equivalent)***

[d-5] FOOD TECHNOLOGY

Vitamins

- Fat-soluble: A (vision/skin), D (Ca absorption/bones), E (antioxidant), K (clotting)
- Water-soluble: B_1(thiamine), B_2(riboflavin), B_{12}(cobalamin), C(ascorbic acid/immunity)

★ Deficiency: A→night blindness, D→rickets, B_1→beri-beri, C→scurvy

Cereals, Pulses & Milk

- Cereals: wheat(gluten), rice(starch), maize, bajra, jowar — main carbohydrate source
- Processing: milling (flour), parboiling (rice), malting (barley for beer/malt)
- Pulses: dal, chickpea, soybean, lentils — 20–25% protein, dietary fibre

■ ***Milk composition: ~87% water, 3.5% fat, 3.5% protein(casein), 4.9% lactose***

- Milk products: butter (churning), cheese (coagulation+curing), ghee (clarified butter)
- Milk powder: spray drying (fine powder, soluble) or roller drying

Pasteurization & Preservation

■ ***HTST: 72°C for 15 seconds | LTLT: 63°C for 30 minutes | UHT: 135°C, 2–4 sec***

- Food preservation methods: Canning (heat+seal), Freezing (–18°C), Dehydration/drying
- Pickling (acid/salt), Vacuum packing, Irradiation (gamma rays), Chemical preservatives
- BIS/FSSAI standards govern food quality and safety in India

[d-6] OTHER ENGINEERING COURSES

d-6(i) Agriculture Engineering

- Tractors: 2WD/4WD; power take-off (PTO) for implements
- Tillage implements: MB plough (primary), cultivator, harrow (secondary)
- Sowing: seed drill (row, uniform depth); transplanter for paddy
- Harvesting: combine harvester (cuts+threshes+cleans in one pass)
- Irrigation equipment: pump sets, drip emitters, sprinkler heads, rain guns
- Power tiller: 2-wheel tractor; suitable for small/marginal farmers

d-6(ii) Architecture

- Indus Valley: grid planning, drainage, burnt bricks (Mohenjodaro, Harappa)
- Mauryan: stone pillars, Ashoka pillars, rock-cut caves (Chandragupta, Ashoka era)
- Gupta: Nagara temple style (North India shikhara)
- Mughal: Persian+Indian blend; Taj Mahal, Red Fort, Jama Masjid
- Colonial: Indo-Saracenic, neoclassical; Victoria Memorial, India Gate
- Building typologies: Residential, Commercial, Institutional, Industrial, Recreational
- Design principles: Form, Function, Structure, Aesthetics, Sustainability, Proportion

✔ Vastu Shastra: traditional Indian spatial arrangement philosophy

d-6(iii) Fashion Design & Technology

- History: Ancient (draped — sari, dhoti) → Medieval (layered) → Renaissance (tailored) → Modern
- Elements: Line, Shape, Colour, Texture, Pattern, Value, Proportion
- Colour wheel: Primary(R/Y/B), Secondary(O/G/V), Complementary(opposite), Analogous(adjacent)
- Garment process: Design sketch → Pattern drafting → Fabric cutting → Stitching → Finishing → QC
- Machines: Industrial lockstitch, Overlock/Serger, Interlock, Buttonhole, Flatlock
- CAD/CAM: Gerber, Optitex, Lectra — pattern grading, marker making, fabric planning
- Fashion weeks: Paris, Milan, New York, London — 4 major global fashion capitals